

CLIENT SUCCESS STORY

Profitero

Product Identity Resolution at Global Scale

80%+

AUTOMATED ACCURACY

300+

CLIENT DATASETS

1000+

DATASET SCALABILITY

— BACKGROUND

The Challenge of Product Identity at Scale

Profitero operates across hundreds of retailers and brands, ingesting massive volumes of product data across geographies, languages, and formats. At this scale, the concept of a “product” is not stable — it is a moving target.

THE REALITY

The same physical product appears with different identifiers, inconsistent attributes, across multiple retailers and languages — as separate SKUs with no deterministic linkage between them.

THE SCALE

300+ client datasets. Multilingual catalogs spanning dozens of markets. Inconsistent and incomplete data sources that defeat any single-method matching strategy.

THE CORE TENSION

“Product identity is not a field. It is an inference problem over incomplete data.”

— PROBLEM

Where Existing Approaches Broke Down

Legacy Approach

- Rule-based matching logic
- Identifier normalization only
- Manual validation workflows

Failure Conditions

- Missing or corrupted identifiers
- High attribute variability
- Cross-retailer inconsistencies

CONSEQUENCES

- × Incomplete and inconsistent product grouping across the catalog
- × High operational overhead from manual review and correction
- × Limited scalability — system could not grow with the business

— SOLUTION

A Multi-Layer Identity Resolution System

Paladio AI designed a five-layer identity resolution architecture combining **deterministic logic**, **statistical similarity modeling**, **semantic vector search**, **LLM-based reasoning**, and **ensemble clustering** — each layer compensating for the limitations of the last.

LAYER 01

Deterministic Identity Resolution

- Normalize UPC / EAN / SKU identifiers
- Establish high-confidence base clusters
- Handle malformed or partial identifiers

LAYER 02

Attribute Similarity Modeling

- Tokenize and normalize product metadata
- Apply similarity scoring across titles, brands, categories
- Recover near-matches beyond exact identifiers

LAYER 03

Semantic Matching

- Generate vector embeddings for all products
- Compare candidates using cosine similarity
- Match across language differences and partial data

LAYER 04

Model-Based Reasoning

- Re-rank candidate matches using LLMs
- Classify duplicate vs. non-duplicate pairs
- Resolve ambiguous edge cases with contextual reasoning

LAYER 05

Ensemble Clustering

- Combine outputs from all upstream layers into a unified signal
- Prioritize high-confidence matches (>0.99 threshold)
- Construct final unified product clusters across all datasets

— ARCHITECTURE

System Design Principles

**API-Driven Orchestration**

Modular, scalable pipeline execution across all data sources via API.

**Continuous Ingestion**

Incremental updates and model retraining as new data arrives.

**Confidence-Gated Review**

Human review limited to low-confidence cases. Every decision is scored.

— RESULTS

MEASURABLE OUTCOMES

Accuracy, Scale, and Speed — Delivered Together

The system achieved what rule-based approaches could not: reliable product identity across hundreds of retailers, dozens of languages, and thousands of datasets.

80%+

Automated deduplication accuracy
across all datasets

1000+

Dataset scalability — far beyond the
original 300



Material reduction in manual validation
effort and latency



Cross-retailer, multilingual product matching at
production scale



Significantly reduced manual validation overhead



Reduced processing latency and faster
turnaround time



Scalable architecture ready for continued data
growth

— IMPACT

From Static Rules to a Dynamic Identity Layer

BEFORE

Static, rule-based matching — brittle,
manual, and unable to adapt to data
variability or scale



AFTER

A dynamic product identity layer capable of
adapting to any data variability, language, or
retailer format

THIS ENABLED

- Faster onboarding of new client datasets
- Improved data consistency across all products
- Reduced operational cost at scale

KEY INSIGHT

Product identity is not a field.

It is an inference problem over incomplete data.